

Project Title: Cross-Sportsbook Arbitrage Detection Using Web-Scraped Betting Odds

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Project Description

This project investigates whether differences in betting odds across multiple sportsbooks can be used to identify arbitrage opportunities. An arbitrage occurs when odds across platforms allow a bettor to guarantee profit regardless of the outcome. The goal is to collect and combine odds data from multiple sources and analyze how often these opportunities appear, which combinations of sportsbooks provide the best likelihood of arbitrage opportunities, and how large the arbitrage opportunities are.

Data Sources

- **DraftKings Sportsbook (Scraped)**

<https://sportsbook.draftkings.com/>

Data will be collected using Selenium due to dynamic content loading, as well as requests to establish connection to the page before scraping is permitted. Extracted fields include teams, market (totals), and odds.

- **BetMGM (Scraped)**

<https://www.az.betmgm.com/en/sports?wm=7119139&btag=&tdpeh=&pid=>

Similar scraping methods will be used to extract comparable betting data for the same events, as events contain hyperlinks much like DraftKings making code easier to reuse.

- **The Odds API (API Source)**

<https://the-odds-api.com/>

This API provides structured JSON data containing odds across multiple bookmakers and will be accessed using Python requests.

Data Integration

This project will focus on totals (over/under), because the line (like 6.5) is usually the same across sportsbooks, which makes it much easier to match the same bet across different sites and compare the odds. In order to combine data accurately, a `canonical_event_id` will be created using teams, market, and line. This allows matching equivalent events across different sportsbooks.

Analysis Plan

Arbitrage opportunities will be identified by converting odds into implied probabilities and checking:

$$\frac{1}{odds_1} + \frac{1}{odds_2} < 1$$

If this condition holds, a risk-free profit opportunity exists. The analysis will focus on:

- Frequency of arbitrage opportunities, visualized using a line graph.
- Which sportsbook pairs produce the most opportunities, visualized using a heat map.
- Distribution of profit margins, visualized using a bar graph.

Research Questions

This project aims to answer:

- How often do arbitrage opportunities occur across sportsbooks?
- Which combinations of sportsbooks are most likely to produce arbitrage?
- What patterns exist in arbitrage opportunities within totals (over/under) markets?